UAV-GAP / CORE RESEARCH EVIDENCE

Two variability spaces • Cross-model interactions • Robustness gaps

01 Validated variability spaces

FM_P | Degradation variability

46,368

25 leaves · 5 constraints · no dead features

→

Valid configurations

→ sampled pairs → YOLO + ML

←

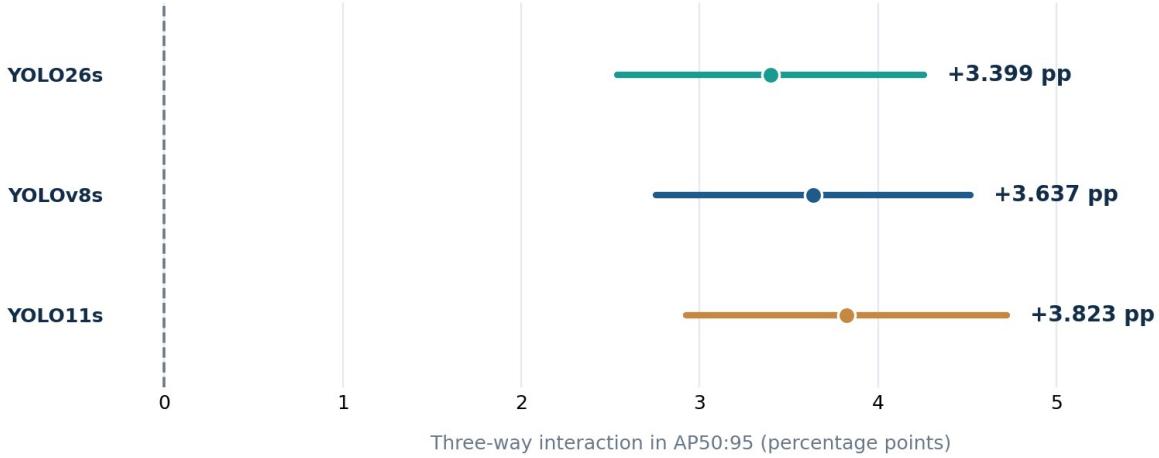
FM_D | YOLO variability

635,904

34 leaves · 4 constraints · 844 training variants

THEORETICAL CROSS-PRODUCT 29,485,596,672 configuration pairs | not an executed evaluation count

02 Interaction: resolution × blur × JPEG | development evidence



KEY FINDING

The effect of higher inference resolution changes with blur and JPEG conditions.

All three reported three-way contrasts have positive 95% simultaneous intervals.

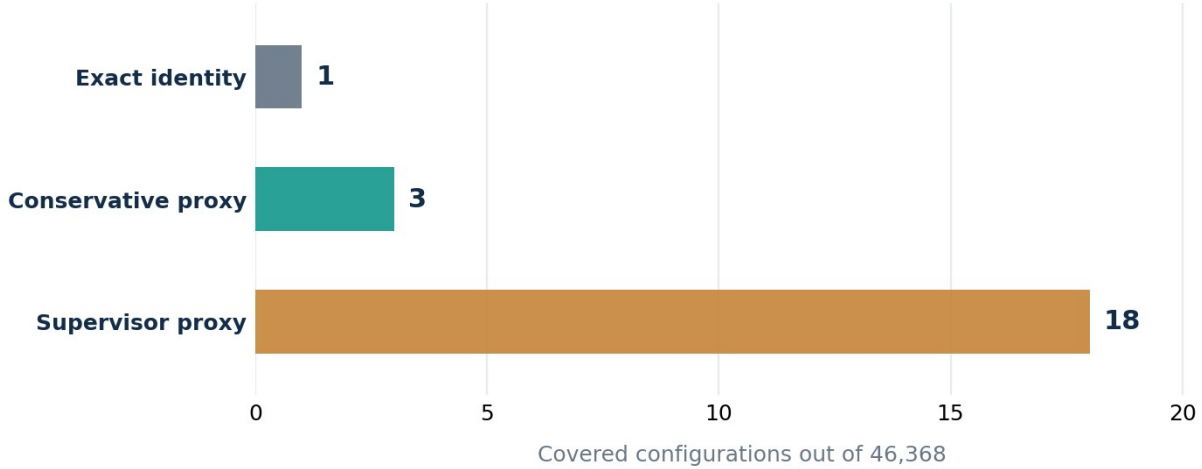
DEVELOPMENT ONLY · No independent held-out ML result.

Source: VM26 / CV261207 accepted historical development evidence; 300 VisDrone val images; confidence intervals from the reported joint bootstrap.

UAV-GAP / CORE RESEARCH EVIDENCE

Static coverage sensitivity • Measured severe-blur failures • Supported conclusions

03 Static gap | augmentation coverage assumptions



STATIC UNCOVERED

46,367 · exact identity

46,365 · conservative proxy

46,350 · supervisor proxy

A static gap is NOT a measured detector-failure rate.

04 Measured Tiny-object failure | evaluated detector subset

6	2	< 0.10
detector settings 3 checkpoints × 2 resolutions	severe-blur conditions B2 × JPEG None / Q50	Tiny AP50:95 upper CI all six settings in both conditions

Conclusion: under the chosen Tiny AP threshold, the six measured settings fail in these two severe-blur conditions; no claim about the full FM_D space.

05 Conclusions and evidence boundary

RQ1	LOGIC VALIDATED	FM_P and FM_D are satisfiable with no dead features; complete detector deployability is not established.
RQ2	DEVELOPMENT EVIDENCE	Resolution benefits are jointly conditioned by blur and JPEG; independent predictive validation remains open.
RQ3	PARTIAL	Static coverage is policy-sensitive; real subset failures are observed, but global empirical gap is unknown.
RQ4	NOT EVALUATED	No gap-closure effect can be claimed without matched targeted-augmentation training.

Evidence scope: historical development data; formal independent test-dev 0/500. CIs are reported source results, not newly recomputed from raw bootstrap samples.